

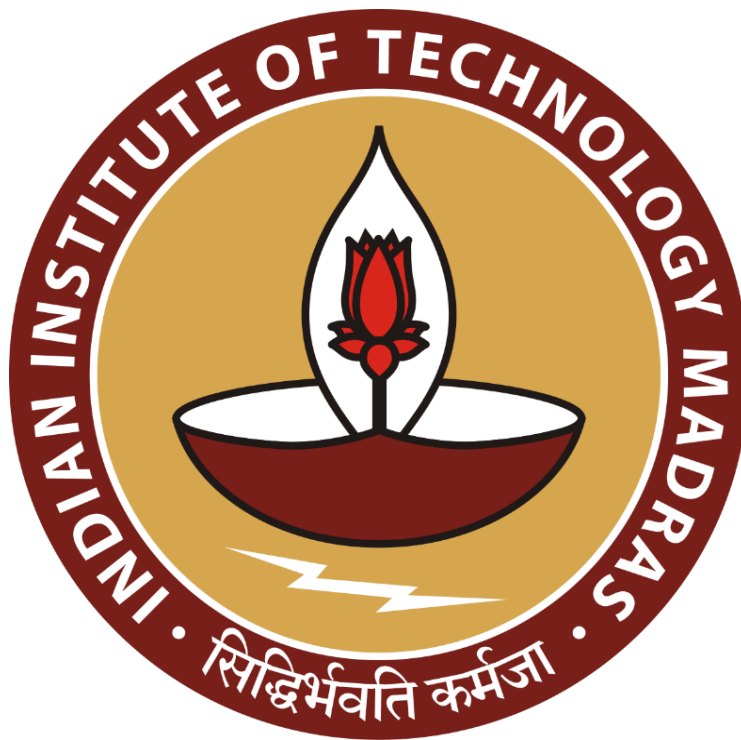
Strategic Growth and Revenue Stabilization: Enhancing Customer Retention and Inventory Management

A mid-term report for BDM Capstone Project

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1. Executive Summary

‘S’ and ‘S’ Brothers, established in 2015, is a well-known wholesale dealer in jaggery based in Kozhikode Big Bazaar. The business operates in both B2B and B2C capacities with more than 8 years of experience. The owner of S and S Brothers brings over 25 years of experience in the jaggery trade. Before starting this independent venture, he was in a partnership business with his older brothers in the same industry. ‘S’ and ‘S’ Brothers offers high-quality, FSSAI-certified jaggery products, ensuring safety and compliance with food standards. The product range includes jaggery cubes, jaggery balls, and powdered jaggery, catering to different customer needs. The company’s revenue has decreased more than half, from 25 lakh over the past five years. The core challenges include revenue instability, inefficient inventory management and outdated marketing strategies.

The data collected consists of detailed sales records and stock ledger information spanning a year. The preliminary analysis has been done on data of six-month time period, with key metadata including columns such as opening stocks, quantity sold, product type, sales rate and stock available. These records provide an initial indication of sales trend, product performance and inventory cycles. Descriptive statistics such as mean- sales per product, revenue turnover and stock trends have been generated to provide insights into operational efficiency.

This project utilizes Python libraries (pandas, NumPy, matplotlib and seaborn) and Excel for data cleansing, visualization and statistical analysis. Initial findings reveal - sales trend, fluctuations in revenue, inconsistencies in stock replenishment. The project aims to optimize revenue and sales trend through time series analysis, regularity in inventory management through ABC analysis and other visualization tools.

2. Proof of Originality

2.1 Letter from organization: [Letter](#)

2.2 Video of Interaction with the Business Owner (in Malayalam): [Video](#)

2.2 Images Related to the organization



Fig 2.2.1 Frontage of the shop with owner.



Fig 2.2.2 Products specified in the data.

Fig 2.2.3 The sales book and stock register provided by the owner.

3. Metadata & Descriptive Statistics

Data collected for this project contains day-wise sales in S and S Brothers in the form of excel sheet for a period of six months.

Data Contents:

Information about Stock Ledger data: [Stock Ledger](#)

Link:

The Header – The header is described by the respective product name and the product type.

Month, Date, Year – The Month, Date, Year of the respective sale.

Opening Stock – The opening stock of that particular product for the respective date.

By Sales – The number of sales issued for that particular product on the respective date.

Incoming Stocks – The number stocks that are newly added for that particular product on the respective date.

Total Stocks – The net total stock remaining after the sales and incoming stock from the opening stock on the respective date.

Revenue By Sales – The total Revenue generated by that particular product on that respective date. It is the product of unit price with the number of sales.

Average Unit Price - The average of the unit prices for a particular product over a month.

Total revenue – The total revenue produced over a month.

Information about Average Price data: [Average price](#)

It contains tables under two headings:

Sales Price data

Month– The respective month of the data,

Average of the month – The average unit price for the respective vendor for the respective month.

Average over 6 months - The Average of the average unit price collected for the respective month over a 6-month time period.

Ratio of Change = $(\text{Average of the month} - \text{Average over 6 months}) / \text{Average of the month} * 100$

It describes the respective change in percentage of average monthly unit price from the 6-month average.

No of sales data

Month– The respective month of the data,

Sales of the month – The respective total sales in the given month.

Average over 6 months - The Average of the sales of the month collected for the respective month over a 6-month time period. `

A brief picture of **preliminary analysis** has also been shared: [Preliminary Analysis](#).

Products – The different products offered.

Active sales – day specifies the number of days on which the particular product was sold.

No sales day – specifies the number of days on which the product type went unsold.

Ratio % – the ratio of total sales to active sales days.

Cumulative revenue – successive addition of Total Revenue by each product.

Justification

By looking at details like Revenue by Sales, Average Unit Price, Total Revenue and Sales of the Month, we can get a clear picture of how income changes from month to month. These values help us understand how stable or unpredictable each product type is sold, which directly relates to recognizing trends in demand and price competitiveness.

Fields like Opening Stock, Incoming Stocks, and Total Stocks offer insights into how well each product category is managed in the inventory. Whether running low too frequently or retaining excess stock,

these figures can identify where the balance is incorrect—something which is integral to addressing the stock management challenges highlighted in the problem statement.

The below data explains the Descriptive-Statistics on the sales price range for each product:

INDEX NO	Products	MEAN	STD DEV	MIN	MAX
1	Bharati Traders - Jaggery nuts	1352.49	90.05	1233.66	1464.32
2	Sri Jaya Bharati Traders - Jaggery nuts	1306.46	103.08	1215.33	1453.24
3	Reshma Traders - Jaggery nuts	1303.25	69.75	1231.62	1428.90
4	Sri Dharani Traders - Jaggery balls	1640.00	119.37	1550.00	1850.00
5	Sri Vengadesh Perumal Traders - Jaggery balls	1457.17	89.42	1400.00	1591.00

Fig 3.1 D-stats table on sales price data.

Descriptive-Stat on the sales data for each product:

INDEX NO	Products	MEAN	STD DEV	MIN	MAX
1	Bharati Traders - Jaggery nuts	376	157	205	615
2	Sri Jaya Bharati Traders - Jaggery nuts	89	82	0	196
3	Reshma Traders - Jaggery nuts	163	61	72	244
4	Sri Dharani Traders - Jaggery balls	138	101	0	260
5	Sri Vengadesh Perumal Traders - Jaggery balls	343	238	193	814

Fig 3.2 D-stats table on number of sales data.

Justification

Standard deviation in number of sales and price helps quantify the level of instability in revenue and market pricing.

Descriptive statistics on the sales data for each product type highlight patterns of under- and over-supply, enabling targeted inventory adjustments.

4 Detailed Explanation of Analysis process

4.1 Collection

The raw data consisted of monthly sales figures for five different product types—Bharathi Traders-Jaggery nuts, Sri Jaya Bharathi Traders-Jaggery nuts, Reshma Traders-Jaggery nuts, Sri Dharani Traders-Jaggery balls, and Sri Vengadesh Perumal Traders-Jaggery balls, from April to September. Initial cleaning was done using Excel, where blank cells and zero entries (e.g., zero sales in July and September for Sri Jaya Bharathi Traders and Sri Dharani Traders) were verified and maintained as significant data points rather than errors. Standardizing month and trader

names, Handling outliers or inconsistencies in average unit prices. The data was subjected to preliminary analysis and visualizations in Excel. The dataset was also imported into Python (using Pandas) for descriptive analysis and visualization.

Cleaning the data was critical to ensure consistency and accuracy across all variables. For example, zero-sales months could skew averages if not treated carefully. Standardization of column headers ensured that automated calculations in Python (like groupings and aggregations) worked correctly.

4.2 Analysis

1. Descriptive Statistics

For each trader, computed:

- Sales Price

Mean, Standard Deviation, Minimum and Maximum.

- Total Sales

Mean, Standard Deviation, Minimum and Maximum.

It identifies variability in monthly sales (e.g., Sri Jaya Bharathi Traders shows volatility), which directly connects to – Revenue Instability.

2. Time Series - Sales Trend Analysis

Using Python (pandas, matplotlib, seaborn) and Excel, Time Series Analysis was performed on the monthly sales and revenue data of each trader over a six-month period.

Used line plots and bar charts to visualize sales trends over the six months for each trader.

Helped identify seasonal patterns, and inconsistent recovery, supporting sales and revenue pattern insights.

Sales trends (rising, falling, or stable)

Seasonal effects (e.g., festive spikes or off-season slumps)

Revenue inconsistencies linked to pricing or supply issues

It directly quantifies how revenue varies over time and identifies months of unusually high or low performance, helping isolate causes like pricing issues, inventory problems, or external events.

Although full time series modeling is planned in future stages, current monthly trends were plotted to show patterns and anomalies.

This forms the base for later predictive tasks like forecasting future revenue.

This also helps in identifying proper marketing strategies to be implemented in the right time.

3. ABC Analysis for Inventory (Using Pareto Chart)

ABC Analysis is an inventory management technique that categorizes items based on their consumption value.

ABC analysis was used on stock data of all five products to categorize them based on sales contribution.

Sales totals were ranked to apply ABC categorization.

Steps: Calculate each product's total value: Unit Price $\times$ Quantity Sold

Sort products in descending order of total value

Compute cumulative percentage contribution. Based on this the top 70% revenue contributing items were ranked as A. The next 20% as B. The last 10% as C

This method is ideal for inventory control, aligning with– Inventory Management Issues. It provides clarity on stock priority for restocking or pricing and enables better allocation of attention and resources toward high-impact items.

5. Results and Findings

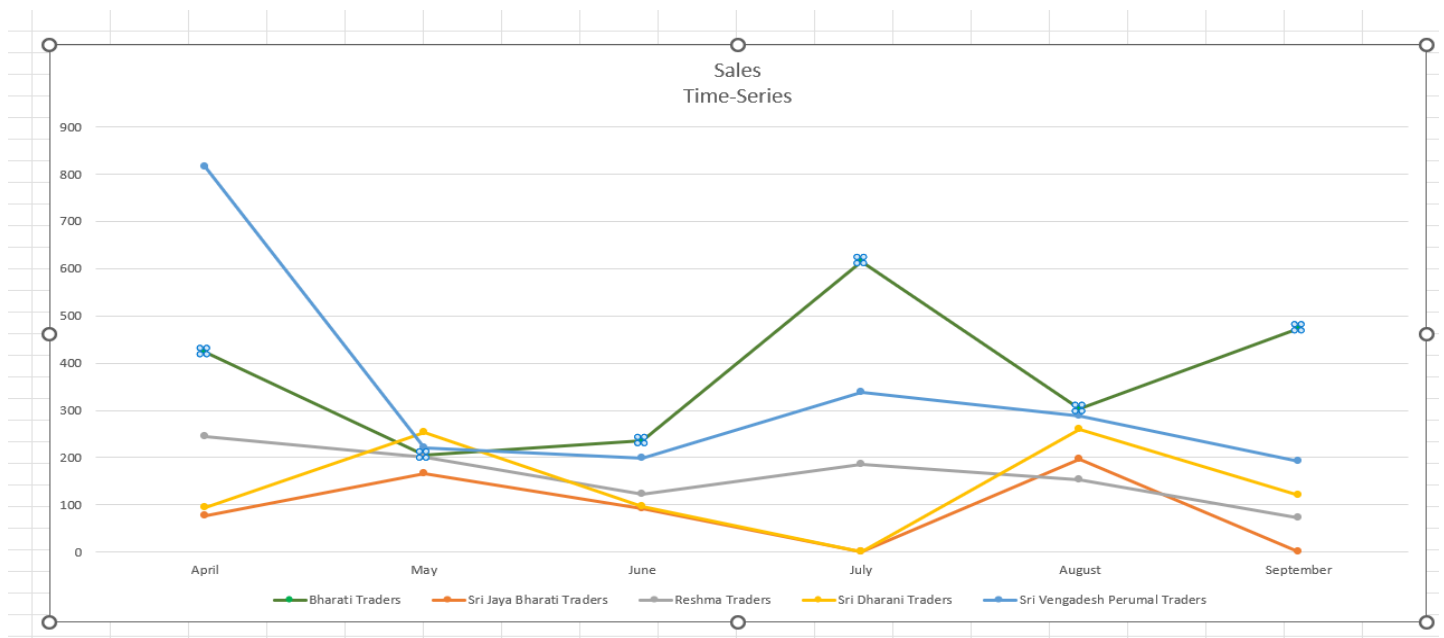


Fig 5.1 Time-Series graph of no of sales of each product.

The above graph provides a time-series plot of sales of different products. The graph shows significant fluctuations in sales across different products and months. For instance, Sri Vengadesh Perumal Traders had a peak in April but a sharp decline by June. Bharati Traders show a sharp increase in July. Such variability in sales can be a direct representation of the revenue instability due to fluctuating customer demand and competitive market pricing.

Another possibility might suggest that they had insufficient stock to meet a sudden increase in demand. This aligns with the problem of stock imbalances.

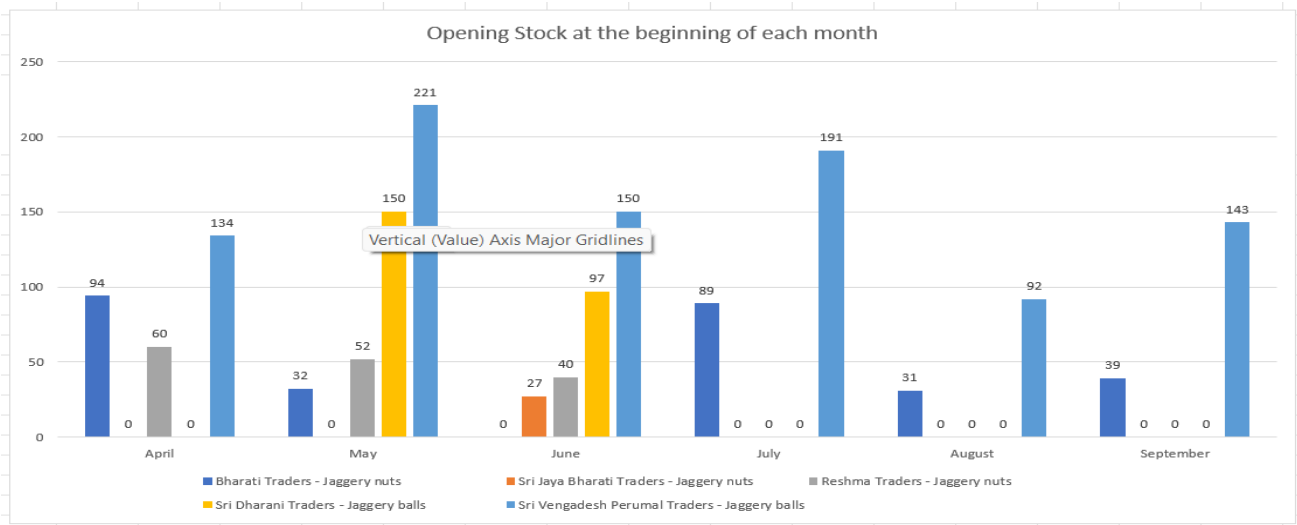


Fig 5.2 Opening stock at the beginning of each month for each product.

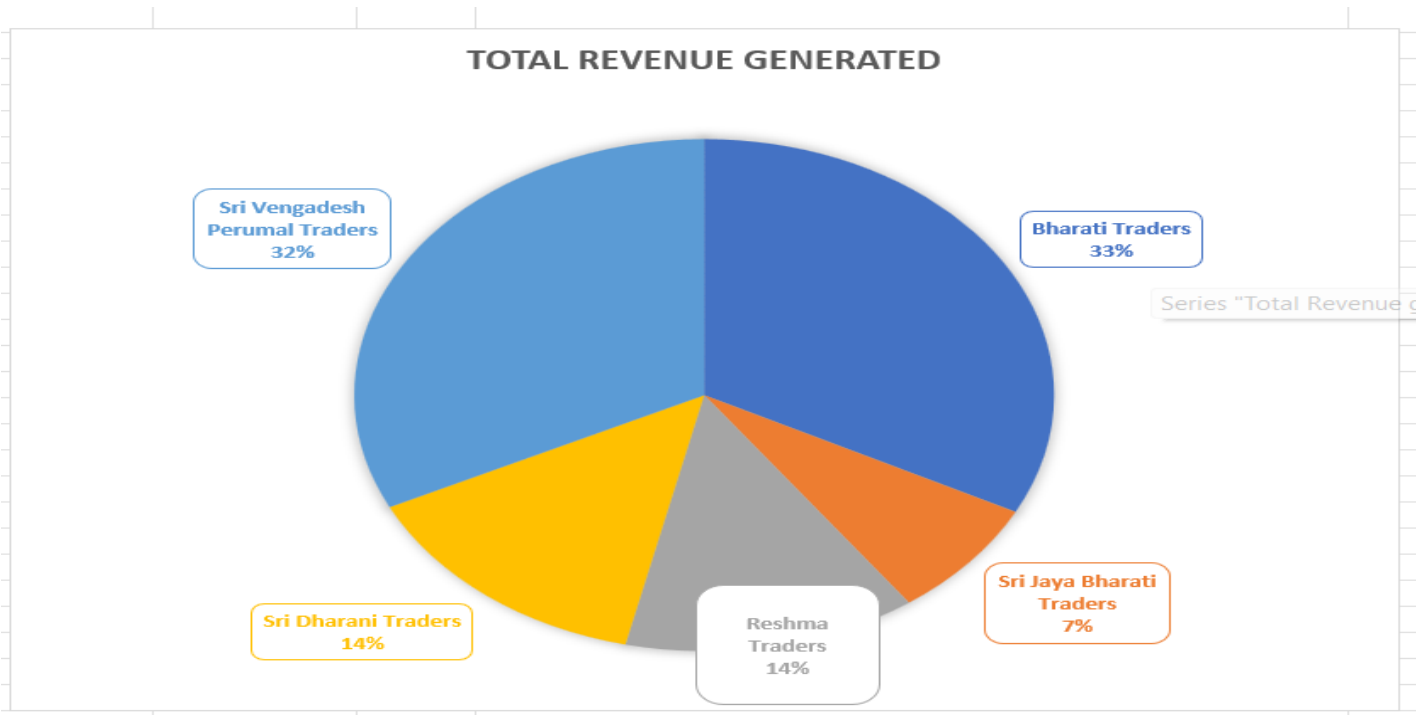


Fig 5.3 Pie Chart representing the revenue share by each product.

Bharati traders-jaggery nuts generates the maximum revenue of ₹30,69,625 contributing 33% of **S and S brothers** revenue and Sri Jaya Bharati-jaggery nuts contributes the least ₹7,08,765 (7%). Products with higher sales volumes (Bharati and Sri Vengadesh Perumal), Analyzing inventory data for these products could help identify patterns and improve inventory management across the board.

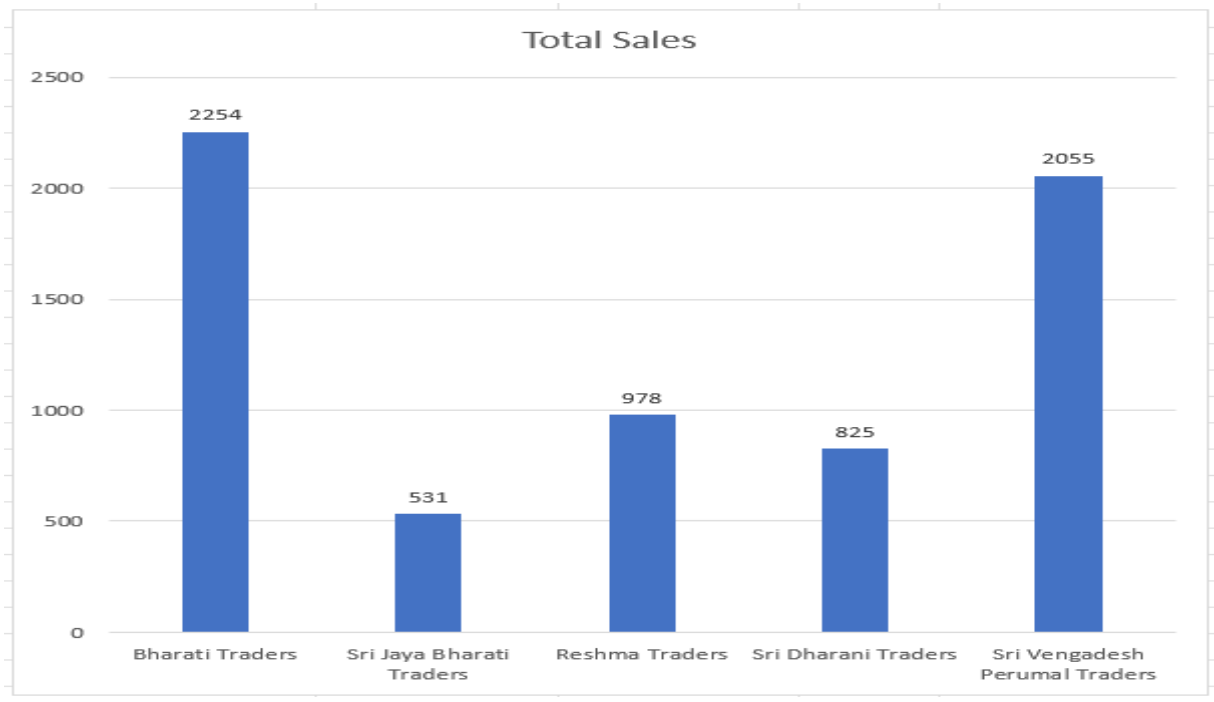


Fig 5.4 Bar chart which represents the total no of sales of each product.

The total sales of Bharati traders and Sri Vengadesh Perumal traders sum up to 2254 and 2055 whereas Sri Jaya Bharati has a total sale of 531.

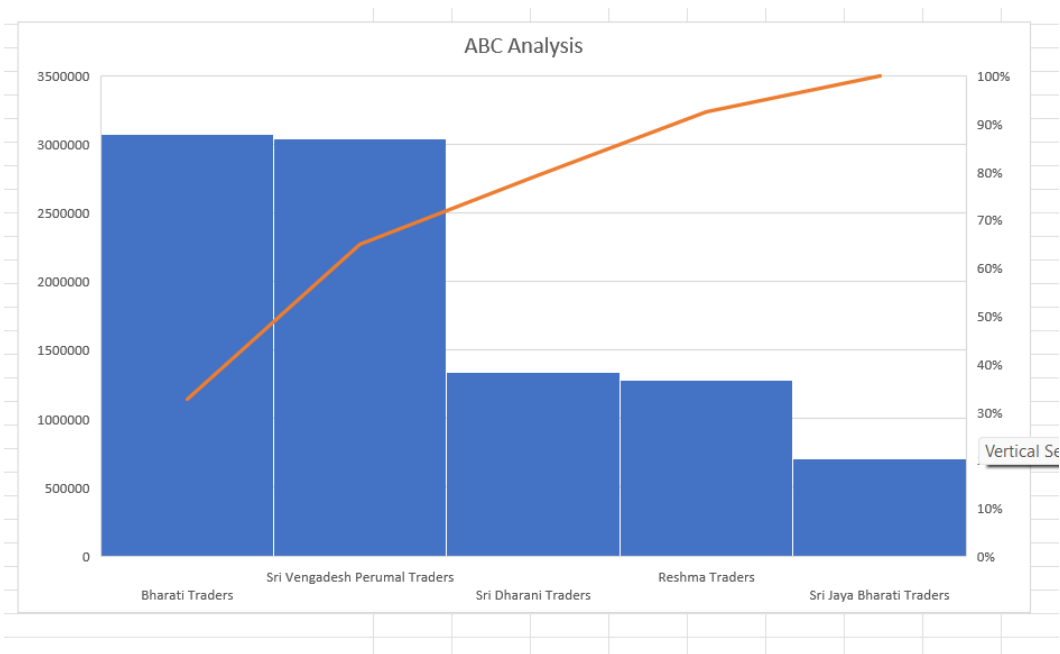


Fig 5.5 Pareto chart used for ABC analysis.

The graph is an ABC analysis chart, which is used to categorize items based on their importance or value.

Bharati Traders-jaggery nuts and **Sri Vengadesh Perumal Traders-jaggery balls** contribute significantly to the total sales, as indicated by their high bars and the steep initial rise in the cumulative percentage line.

Sri Dharani Traders-jaggery balls and **Reshma Traders-jaggery nuts** contribute less but still significantly to the total sales.

Sri Jaya Bharati Traders-jaggery nuts contribute the least to the total sales.



Fig 5.6 Marked Line Chart for sales trend and pricing trend.

The above graph shows the deviation of Sales price of each month to the average of sales month over 6 months. It shows a specific trend for each month for each product. The above graph only provides the trend for Sri Vengadesh Perumal – jaggery balls, The sales volume is highly variable, with a sharp decrease in May and a gradual recovery through August. The slight decrease in September could indicate a seasonal downturn or other market factors affecting sales.

The fluctuation in both price and sales volume contributes to revenue instability. The sharp drop in sales in May and the significant price increase in September could both impact revenue unpredictably.

The variability in sales could also be influenced by the lack of a structured marketing strategy. The sharp drop in May and the recovery thereafter might indicate the impact of ad-hoc marketing efforts or seasonal factors rather than a consistent marketing strategy.

While the graph does not directly show quality issues, inconsistent quality could lead to fluctuations in sales volume as customers might switch to competitors if they perceive a decline in product quality.